

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

If $5f(x) + 4f\left(\frac{1}{x}\right) = x^2 - 2, \forall x \neq 0$ and $y = 9x^2f(x)$, then y is strictly increasing in :

- (1) $\left(0, \frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$
- (2) $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$
- (3) $\left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$
- (4) $\left(-\infty, \frac{1}{\sqrt{5}}\right) \cup \left(0, \frac{1}{\sqrt{5}}\right)$

Q2 - 2024 (27 Jan Shift 2)

Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for all $x \in (0, 3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is

- (1) 24
- (2) 0
- (3) 18
- (4) 20

Q3 - 2024 (29 Jan Shift 2)

The function $f(x) = 2x + 3(x)^{\frac{2}{3}}, x \in \mathbb{R}$, has

- (1) exactly one point of local minima and no point of local maxima
- (2) exactly one point of local maxima and no point of local minima
- (3) exactly one point of local maxima and exactly one point of local minima
- (4) exactly two points of local maxima and exactly one point of local minima

Q4 - 2024 (29 Jan Shift 2)

The function $f(x) = \frac{x}{x^2 - 6x - 16}, x \in \mathbb{R} - \{-2, 8\}$

- (1) decreases in $(-2, 8)$ and increases in $(-\infty, -2) \cup (8, \infty)$

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(2) decreases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ (3) decreases in $(-\infty, -2)$ and increases in $(8, \infty)$ (4) increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

Q5 - 2024 (30 Jan Shift 1)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a non constant twice differentiable such that $g' \left(\frac{1}{2} \right) = g' \left(\frac{3}{2} \right)$. If a real valued function f is defined as $f(x) = \frac{1}{2}[g(x) + g(2-x)]$, then

(1) $f''(x) = 0$ for atleast two x in $(0, 2)$ (2) $f''(x) = 0$ for exactly one x in $(0, 1)$ (3) $f''(x) = 0$ for no x in $(0, 1)$ (4) $f' \left(\frac{3}{2} \right) + f' \left(\frac{1}{2} \right) = 1$

Q6 - 2024 (30 Jan Shift 2)

Let $f(x) = (x+3)^2(x-2)^3, x \in [-4, 4]$. If M and m are the maximum and minimum values of f , respectively in $[-4, 4]$, then the value of $M - m$ is :

(1) 600

(2) 392

(3) 608

(4) 108

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Q5 (1)  mathongo  mathongo **Q6** (3)  mathongo  mathongo  mathongo  mathongo

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Solutions

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Q1

$$5f(x) + 4f\left(\frac{1}{x}\right) = x^2 - 2, \forall x \neq 0$$

Substitute $x \rightarrow \frac{1}{x}$

$$5f\left(\frac{1}{x}\right) + 4f(x) = \frac{1}{x^2} - 2$$

On solving (1) and (2)

$$f(x) = \frac{5x^4 - 2x^2 - 4}{9x^2}$$

$$y = 9x^2 f(x)$$

$$y = 5x^4 - 2x^2 - 4 \dots (3)$$

$$\frac{dy}{dx} = 20x^3 - 4x$$

for strictly increasing

$$\frac{dy}{dx} > 0$$

$$4x(5x^2 - 1) > 0$$

$$x \in \left(-\frac{1}{\sqrt{5}}, 0\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$$

Q2

$$g(x) = 3f\left(\frac{x}{3}\right) + f(3-x) \text{ and } f''(x) > 0 \forall x \in (0, 3) \Rightarrow f'(x) \text{ is increasing function}$$

$$g'(x) = 3 \times \frac{1}{3} \cdot f'\left(\frac{x}{3}\right) - f'(3-x)$$

$$= f'\left(\frac{x}{3}\right) - f'(3-x)$$

If g is decreasing in $(0, \alpha)$

$$g'(x) < 0$$

$$f'\left(\frac{x}{3}\right) - f'(3-x) < 0$$

$$f'\left(\frac{x}{3}\right) < f'(3-x)$$

$$\Rightarrow \frac{x}{3} < 3-x$$

$$\Rightarrow x < \frac{9}{4}$$

$$\text{Therefore } \alpha = \frac{9}{4}$$

$$\text{Then } 8\alpha = 8 \times \frac{9}{4} = 18$$

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Solutions

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Q3

$$f(x) = 2x + 3(x)^{\frac{2}{3}}$$

$$f'(x) = 2 + 2x^{-\frac{1}{3}}$$

$$= 2 \left(1 + \frac{1}{x^{\frac{1}{3}}} \right)$$

$$= 2 \left(\frac{x^{\frac{1}{3}} + 1}{x^{\frac{1}{3}}} \right)$$



So, maxima (M) at $x = -1$ & minima (m) at $x = 0$

Q4

$$f(x) = \frac{x}{x^2 - 6x - 16}$$

Now,

$$f'(x) = \frac{-(x^2 + 16)}{(x^2 - 6x - 16)^2}$$

$$f'(x) < 0$$

Thus $f(x)$ is decreasing in

$$(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$$

Q5

$$f'(x) = \frac{g'(x) - g'(2-x)}{2}, f'\left(\frac{3}{2}\right) = \frac{g'\left(\frac{3}{2}\right) - g'\left(\frac{1}{2}\right)}{2} = 0$$

$$\text{Also } f'\left(\frac{1}{2}\right) = \frac{g'\left(\frac{1}{2}\right) - g'\left(\frac{3}{2}\right)}{2} = 0, f'\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow f'\left(\frac{3}{2}\right) = f'\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow \text{roots in } \left(\frac{1}{2}, 1\right) \text{ and } \left(1, \frac{3}{2}\right)$$

$$\Rightarrow f''(x) \text{ is zero at least twice in } \left(\frac{1}{2}, \frac{3}{2}\right)$$

Q6

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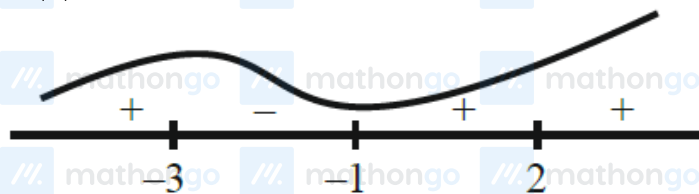
Solutions

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$$f'(x) = (x+3)^2 \cdot 3(x-2)^2 + (x-2)^3 2(x+3)$$

$$= 5(x+3)(x-2)^2(x+1)$$

$$f'(x) = 0, x = -3, -1, 2$$



$$f(-4) = -216$$

$$f(-3) = 0, f(4) = 49 \times 8 = 392$$

$$M = 392, m = -216$$

$$M - m = 392 + 216 = 608$$

$$\text{Ans} = '3'$$

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